BUILT — Look-Alike Parts recognizer (v0.43.0)

Tells apart parts that look identical in the manual but are functionally different — by NSN, UOC, CAGEC, SMR or supply class — with grounded 'how to tell them apart' cues. Read-only.

1 · HOW IT WORKS (/partdiff · /api/partdiff)

1. **Query** — an NSN or a part name
2. **Same-name group** — every catalogued part sharing that name/nomenclature
3. **Group by NSN** — collapse to distinct stock numbers + their attributes
4. **Find discriminators** — which fields vary: NSN · FSC · UOC · CAGEC · SMR · part#
5. **Classify + cross-ref** — vs the correlations sidecar (true substitutes vs real differences)
6. **Compare + cite** — side-by-side cards, how-to-tell-apart, links to the real figure

2 · THE FOUR VERDICTS (colour-coded in the UI)

reference	the part you searched — the anchor for the comparison
different variant	same name, different NSN — usually a different VEHICLE CONFIG. The tell is the UOC (Usable-On-Code); also CAGEC / part#. THIS is the dangerous look-alike.
same item (format drift)	same NIIN, just a different NSN format — genuinely the same part. Interchangeable.
different item class	different FSC — merely shares a figure title (e.g. a bracket inside an 'engine assembly' figure). NOT a substitute.

WHY UOC IS THE KEY

In an RPSTL, two parts can be drawn the same and share a name, yet one fits configuration A and the other configuration B. The Usable-On-Code (UOC) is the field that disambiguates them — pick the wrong NSN and the part may not fit. The recognizer surfaces the UOC contrast first, then CAGEC (maker) and part number.

GROUNDED & ADDITIVE (R1/R6)

Read-only over the existing parts index + the optional correlations sidecar — no new data invented, nothing written to the index. Every variant cites the real figure & page so the mechanic confirms on the sheet. Confirmed-same items (NIIN drift) and cross-platform interchangeable NSNs are labelled as substitutes, not differences. As OCR coverage climbs, item names sharpen and the look-alike groups get finer. New routes only (rollback = remove /partdiff + /api/partdiff). The empty part_variants table is now meaningfully populated on demand at query time.

BUILT diagram. Dark (R3). v0.43.0 · 2026-06-02 · engine/ui/partdiff.html + part_differences() in viewer_app.py. Read-only; correlations sidecar optional.